

Organ shape synthesis using octrees

Weekly Report 10

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What has been done this week

- Writing VAE section and learnt lessons into report
- Implemented some diffusion evaluation metrics
- Evaluated the myocardium diffusion model

Naive diffusion model evaluation

We've implemented three key metrics, being Minimum Matching Distance (MMD), Coverage (COV) and 1-Nearest Neighbour Accuracy (1-NNA), whose purposes vary and whose collected values give a very good estimation of how well a generative model is performing. They are described in the following.

Diffusion metrics

In order to evaluate the quality and diversity of generated meshes systematically, we utilize three commonly used metrics for mesh generation which are explained in the following section. All metrics are distance-based and as such are based on CD and EMD.

Coverage (COV): In order to measure the diversity of the generated meshes we evaluate the coverage as compared to the train/val/test?-set. Let A denote the set of meshes present in $\mathcal{D}_{train}$ of length N with $\tilde{A} = \{\mathbf{X}_i\}_{i=1}^{i=N}$ representing the set of undersampled point cloud from the mesh surface. Likewise, let $\mathcal{D}_{gen}$ be the generated data-set, with meshes B and $\tilde{B} = \{\mathbf{Y}_i\}_{i=1}^{i=M}$ denoting randomly sampled points from the mesh surfaces. For each set element in $\tilde{A}$ we find the closest neighbor in $\tilde{B}$ as measured by CD or EMD. The distance measure defines the neighbourhood-function $\text{NND}_1(\mathbf{Y}_j; \mathbf{X}) = \arg \min_i \|\mathbf{Y}_j - \mathbf{X}_i\|$, where $\|\cdot\|$ is the chosen norm. Then, the coverage is calculated as:

$$\text{COV} = \frac{1}{N} \sum_{i=1}^N \mathbb{1}_{(\exists j \in \{1, \dots, M\} \text{ s.t. } \text{NND}_1(\mathbf{Y}_j; \mathbf{X})=i)} \quad (1)$$

I.e. the coverage measures the fraction of ground-truth samples $\tilde{A}$ which have been matched at least once with a generated sample in $\tilde{B}$ as measured by the neighbourhood-function, and thus gives a measure of the diversity of generated samples[1]. A high coverage is considered better. Note that if $M \leq N$, coverage can maximally attain the value $\frac{M}{N}$.

Minimum Matching Distance (MMD): The coverage metric only tries to measure diversity. A match between two samples may be made which has a large distance, as distance is by no means penalized. As a result, we use MMD to measure how close generated samples $\tilde{B}$ are to samples in $\tilde{A}$. In order to do so, we use the neighbourhood-function $\text{NND}_1(\mathbf{X}_i; \mathbf{Y}) = \arg \min_j \|\mathbf{X}_i - \mathbf{Y}_j\|$ and thus match every generated sample to a single ground-truth sample and report the mean of the minimal distance achieved D :

$$\text{MMD} = \frac{1}{N} \sum_{i=1}^N D(\mathbf{X}_i, \text{NND}_1(\mathbf{X}_i; \mathbf{Y}_{\text{NND}_1(\mathbf{X}_i; \mathbf{Y})})) \quad (2)$$

where D denotes the chosen distance measure $\|\cdot\|$ for the minimal found distance. Thus, MMD may be considered a fidelity-metric, quantifying the average quality of generated meshes[2]. A low MMD is considered better.

1-Nearest-Neighbor Accuracy (1-NNA): The goal of generative modeling is to generate data which is indistinguishable from ground-truth data. This may be quantified using the 1-NNA metric which we use to quantify how distinguishable $\mathcal{D}_{gen}$ is from $\mathcal{D}_{train}$. In order to do so, we construct class-labels c corresponding to the dataset a sample is from, and followingly construct $\tilde{C} = \tilde{A} \cup \tilde{B}$.

We then measure 1-NNA using a metric $\|\cdot\|$ of choice as follows:

$$\text{1-NNA}(\tilde{B}, \tilde{A}) = \frac{\sum_{\mathbf{Y} \in \tilde{B}} \mathbb{1}_{(N_Y \in \tilde{B})} + \sum_{\mathbf{X} \in \tilde{A}} \mathbb{1}_{(N_X \in \tilde{A})}}{|\tilde{B}| + |\tilde{A}|} \quad (3)$$

where N_Y is the 1-nearest neighbourhood-function returning the set-label for the nearest neighbour pointcloud as measured by $\|\cdot\|$ to a sample $\mathbf{Y}$, and likewise for $\mathbf{X}$ [3].

The optimal generative model as measured by 1-NNA would achieve a classification accuracy of 0.5, corresponding to the classifier randomly guessing, ie. having no information present for classification.

Results

Last week, we were very surprised at how well our naive model was performing. Seemingly, it was able to generate high quality synthetic data with little to no flaws, which invoked some skepticism. Generative models often induce undesired behaviour in the form of mode collapse, where the model conforms to learn a small subset of train samples – and in some instances onto a single sample – by heart. This behaviour is undesired, as the model inadvertently learns to deceive the observer, as it is seemingly producing sound generative output, whilst in reality, it has just learned to reproduce training samples with minor variations.

Initially, we performed a test to see whether the same noise-sample $\mathbf{Z}_T$ produces different outputs, which it does. This shows that the sample $\mathbf{Z}_T$ does not map close-to-deterministically to the output mesh. Possibly, one should force determinism to a lower timestep $t < T$.

They may be inspected in Figure 1

In the following figures we present the UMAP for the deterministic samples $\mathbf{Z}_T$

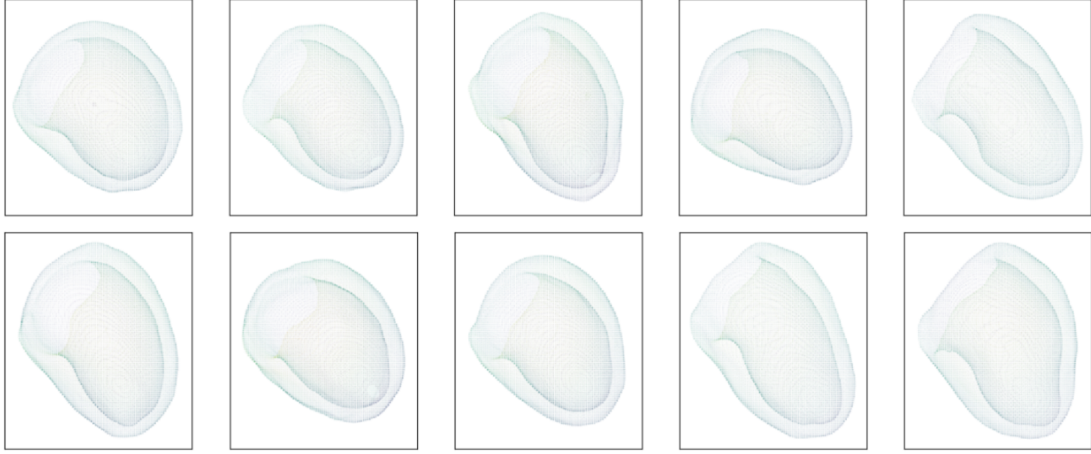


Figure 1: Generated samples using the same $\mathbf{Z}_T$

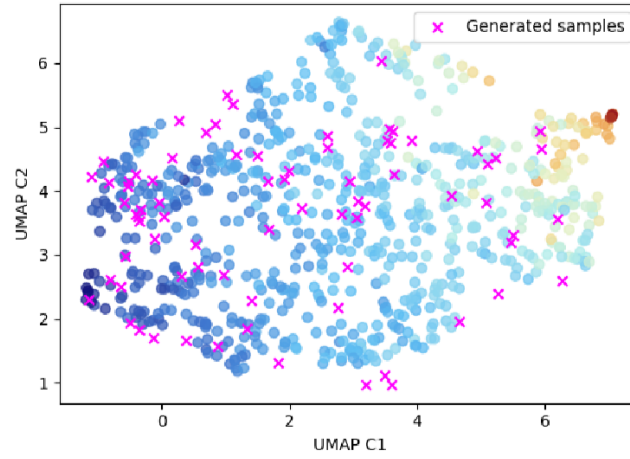


Figure 2: UMAP projection of generated posterior samples $\mathbf{Z}_0$ from a deterministic noise-sample (x's) and posterior samples of the training- and validation sets. UMAP is fitted on train-val encodings.

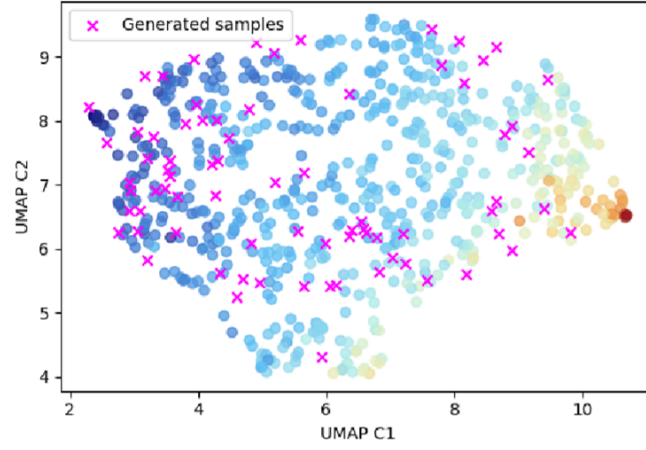


Figure 3: UMAP projection of generated posterior samples $\mathbf{Z}_0$ from a deterministic noise-sample (x's) and posterior samples of the training set. UMAP is fitted on train encodings.

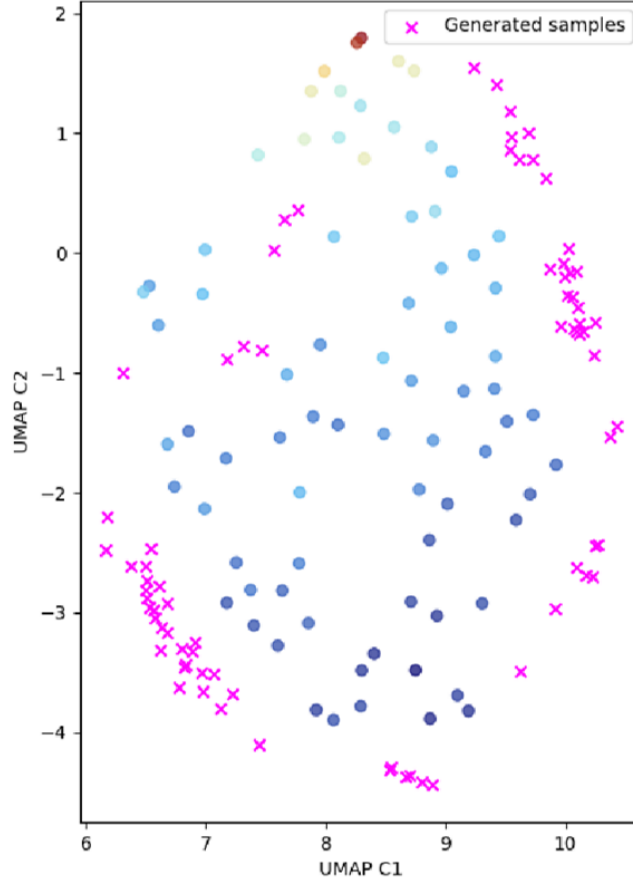


Figure 4: UMAP projection of generated posterior samples $\mathbf{Z}_0$ from a deterministic noise-sample (x's) and posterior samples of the validation set. Most samples look out of distribution of the val-set, but interpreting such a map is hard. UMAP is fitted on val encodings.

Having performed these tests, there were still several loose ends, as one has to be careful drawing too drastic conclusions over the output of a UMAP over the posterior samples, especially as we're not able to confidently insert any KL-threshold, as we've done when evaluating the performance of the VAE. To gain more insights into how well our model is performing, we utilized the aforementioned set of evaluation metrics, MMD, COV and 1-NNA, on our validation set, to see how well our generated output is able to fit to unseen data. The test results are found below:

- lgan_mmd-CD: 6.954058286707951
 - MMD from reference samples to generated samples
- lgan_mmd_smp-CD: 5.740428925123711
 - MMD from reference samples to generated samples
- lgan_cov-CD: 0.4868420958518982
 - Coverage (percentage of reference set covered by generated set)
- 1-NNA-CD-acc: 0.5526315569877625

- Accuracy of 1-NNA (how well does the generated set mix with the reference set)

Producing the same scores over our training set, we find the following:

- lgan_mmd-CD: 6.424346546197666
- lgan_mmd_smp-CD: 2.7173638164384957
- lgan_cov-CD: 0.10947712510824203
- 1-NN-CD-acc: 0.6845930218696594

The found values are quite good compared to as what we expected. Still, we need to check it more thoroughly and consider adding Maximum Mean Discrepancy (also MMD) to figure out if the model is simply down/upscaling meshes.

Questions

- Do you have distributions of features in the ground-truth set already or should we compute them ourselves? Would be nice.
 - We considered fitting an ellipsoid to the outer boundary of round organs, and use ellipsoid parameters as measures of generated samples. Last week we talked about sampling-invariance; would this be sampling invariant "enough"?
- We are discussing whether to evaluate on the very small val-split, on the test-split directly or on the train-split. The train-split is expensive to evaluate upon, and only marginally informative. Test-split evaluation would allow us to incorporate the val-set which would be nice, but bad statistics. In an ideal world we guess we would do train and also val but separately.
- We haven't finetuned this diffusion model, it simply "worked". Is that a fair-enough baseline to compare to down the line? It is the easiest class (myocardium), so bias in missing tuning is maybe alleviated by ease of task.
- Do you have any tips on visualizing many meshes programatically and quickly? Is there any approach which is preferable?

Progress as according to plan

We've spent a lot of time writing and in addition the diffusion model is performing somewhat well, but we still need to figure out if model is simply scaling meshes. This can be measured by maximum mean discrepancy. Thus, we are behind plan, and will have to make a lot of diffusion next week.

Plan for next week

- Build multi-organ support, train on multi-organ VAE?
- Or prioritize: building more complicated diffusion model
- Implement Maximum Mean Discrepancy and check results. This should catch simple scale-variations.
- Conduct a thorough literary review for an optimal diffusion architecture for next model iteration (octfusion, XCube and other candidates)
- Implement realistic augmentations - scaling, rotation, possibly noising.
- Build a pipeline for feature extraction on generated meshes (volumes, euler characteristics(?), fitting an ellipsoid and measuring ellipsoid parameters (but are these sampling-invariant?))

references/references

Bibliography

- [1] Muhammad Ferjad Naeem et al. “Reliable Fidelity and Diversity Metrics for Generative Models”. In: *Proceedings of the 37th International Conference on Machine Learning*. Ed. by Hal Daumé III and Aarti Singh. Vol. 119. Proceedings of Machine Learning Research. PMLR, 13–18 Jul 2020, pp. 7176–7185. URL: <https://proceedings.mlr.press/v119/naeem20a.html> (cit. on p. 1).
- [2] Panos Achlioptas et al. “Learning representations and generative models for 3d point clouds”. In: *International conference on machine learning*. PMLR. 2018, pp. 40–49 (cit. on p. 2).
- [3] Xuanchi Ren et al. “XCube: Large-Scale 3D Generative Modeling using Sparse Voxel Hierarchies”. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2024 (cit. on p. 2).